Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

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TE - 1 Batch: T13 Sem: 5	**Roll No: 2401054**	**Date: 31/07/26**

Experiment 03

Aim: Multi-Service Web Application using Docker Compose

Theory:

Docker

Docker is a containerization platform used to package an application along with its dependencies into a lightweight, isolated environment called a container. Containers ensure that an application runs consistently across different systems.

Docker Image

A Docker image is a read-only template used to create containers. It contains the application code, libraries, dependencies, and configuration required to run an application.

Docker Container

A container is a running instance of a Docker image. It provides an isolated environment in which an application or service can execute.

Dockerfile

A Dockerfile is a text file containing instructions used to build a custom Docker image. In this experiment, the Dockerfile is used to create an image for the Flask backend.

Docker Compose

Docker Compose is a tool used to define and run multi-container applications. Instead of starting every container separately, all services and their configurations can be defined in a single docker-compose.yml file.

In this experiment, Docker Compose manages four services:

Service	**Purpose**
Frontend	Provides the web interface using Nginx
Backend	Runs the Flask application

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PostgreSQL	Stores application data
Redis	Provides caching and maintains the visit counter

Commands used:

Sr. No.	**Command**	**Purpose**
1	mkdir docker-compose-demo	Creates the main project directory
2	cd docker-compose-demo	Navigates to the project directory
3	mkdir backend	Creates the backend directory
4	mkdir frontend	Creates the frontend directory
5	cd backend	Navigates to the backend directory
6	notepad app.py	Creates/opens the Flask application file
7	notepad requirements.txt	Creates the Python dependency file
8	notepad Dockerfile	Creates the Dockerfile for the backend
9	cd ..	Moves to the parent directory
10	cd frontend	Navigates to the frontend directory
11	notepad index.html	Creates the frontend HTML file
12	notepad docker-compose.yml	Creates/edits the Docker Compose configuration
13	notepad nginx.conf	Creates the Nginx configuration
14	dir	Displays files and folders in the current directory
15	tree /f	Displays the complete project directory structure
16	ren Dockerfile.txt Dockerfile	Renames the incorrectly created Dockerfile

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17	docker compose config	Validates the Docker Compose configuration
18	docker compose build	Builds the Docker images
19	docker compose up	Creates and starts all services
20	docker compose up -d	Starts services in detached/background mode
21	docker compose ps	Displays the status of Compose services
22	docker compose down	Stops and removes the Compose containers and network
23	curl http://localhost:5000/health	Tests whether the Flask backend is accessible
24	docker compose logs	Displays service logs

Output:

docker-compose.yml:

```
services:


  frontend:
    image: nginx:alpine
    ports:
      - "8081:80"
    volumes:
      - ./frontend:/usr/share/nginx/html:ro
      - ./nginx.conf:/etc/nginx/conf.d/default.conf:ro
    depends_on:
```

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```
    - web


web:
  build: ./backend
  ports:
    - "5000:5000"
  environment:
    DB_HOST: db
    REDIS_HOST: redis
  depends_on:
    db:
      condition: service_healthy
    redis:
      condition: service_started


db:
  image: postgres:16
  environment:
    POSTGRES_DB: demo_db
    POSTGRES_USER: postgres
    POSTGRES_PASSWORD: password
  healthcheck:
```

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```
    test: ["CMD-SHELL", "pg_isready -U postgres -d demo_db"]
    interval: 5s
    timeout: 5s
    retries: 10

  redis:
    image: redis:7
```

index.html:

```
<!DOCTYPE html>
<html>
<head>
    <title>Docker Compose Demo</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background: #f2f4f7;
            text-align: center;
            padding-top: 70px;
        }
```

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```
.container {
    background: white;
    width: 500px;
    margin: auto;
    padding: 35px;
    border-radius: 12px;
    box-shadow: 0 4px 15px rgba(0,0,0,0.15);
}

h1 {
    color: #333;
}

button {
    padding: 12px 25px;
    font-size: 16px;
    cursor: pointer;
    border: none;
    border-radius: 6px;
    background: #333;
    color: white;
}
```

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```
        #result {
            margin-top: 25px;
            line-height: 1.8;
        }
    </style>
</head>

<body>

<div class="container">

    <h1>Docker Compose Demo</h1>

    <p>Frontend + Flask + PostgreSQL + Redis</p>

    <button onclick="checkBackend()">
        Check Backend
    </button>

    <div id="result">
        Click the button to test the services.
```

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```
    </div>


</div>


<script>
async function checkBackend() {


    const result = document.getElementById("result");


    result.innerHTML = "Checking services...";


    try {


        const response = await fetch("/api");


        const data = await response.json();


        if (data.error) {
            result.innerHTML = "Error: " + data.error;
            return;
        }
```

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```
        result.innerHTML =
            "Message: " + data.message + "<br>" +
            "Database: " + data.database + "<br>" +
            "Cache: " + data.cache + "<br>" +
            "Visits: " + data.visits;

    } catch (error) {

        result.innerHTML =
            "Backend is not reachable.";

    }
}
</script>

</body>
</html>
```

Requirements.txt

```
Flask
psycopg2-binary
redis
```

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app.py:

```
from flask import Flask, jsonify
import psycopg2
import redis
import os

app = Flask(__name__)

DB_HOST = os.getenv("DB_HOST", "db")
REDIS_HOST = os.getenv("REDIS_HOST", "redis")

def get_db():
    return psycopg2.connect(
        host=DB_HOST,
        database="demo_db",
        user="postgres",
        password="password"
    )

redis_client = redis.Redis(
    host=REDIS_HOST,
```

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```
    port=6379,
    decode_responses=True
)

@app.route("/api")
def home():
    try:
        conn = get_db()
        conn.close()

        redis_client.incr("visits")
        visits = redis_client.get("visits")

        return jsonify({
            "message": "Docker Compose application is working!",
            "database": "PostgreSQL connected",
            "cache": "Redis connected",
            "visits": visits
        })

    except Exception as e:
        return jsonify({
```

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```
            "error": str(e)
        }), 500



@app.route("/health")
def health():
    return jsonify({"status": "healthy"})



if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)


nginx.conf:
server {
    listen 80;

    location / {
        root /usr/share/nginx/html;
        index index.html;
    }

    location /api {
```

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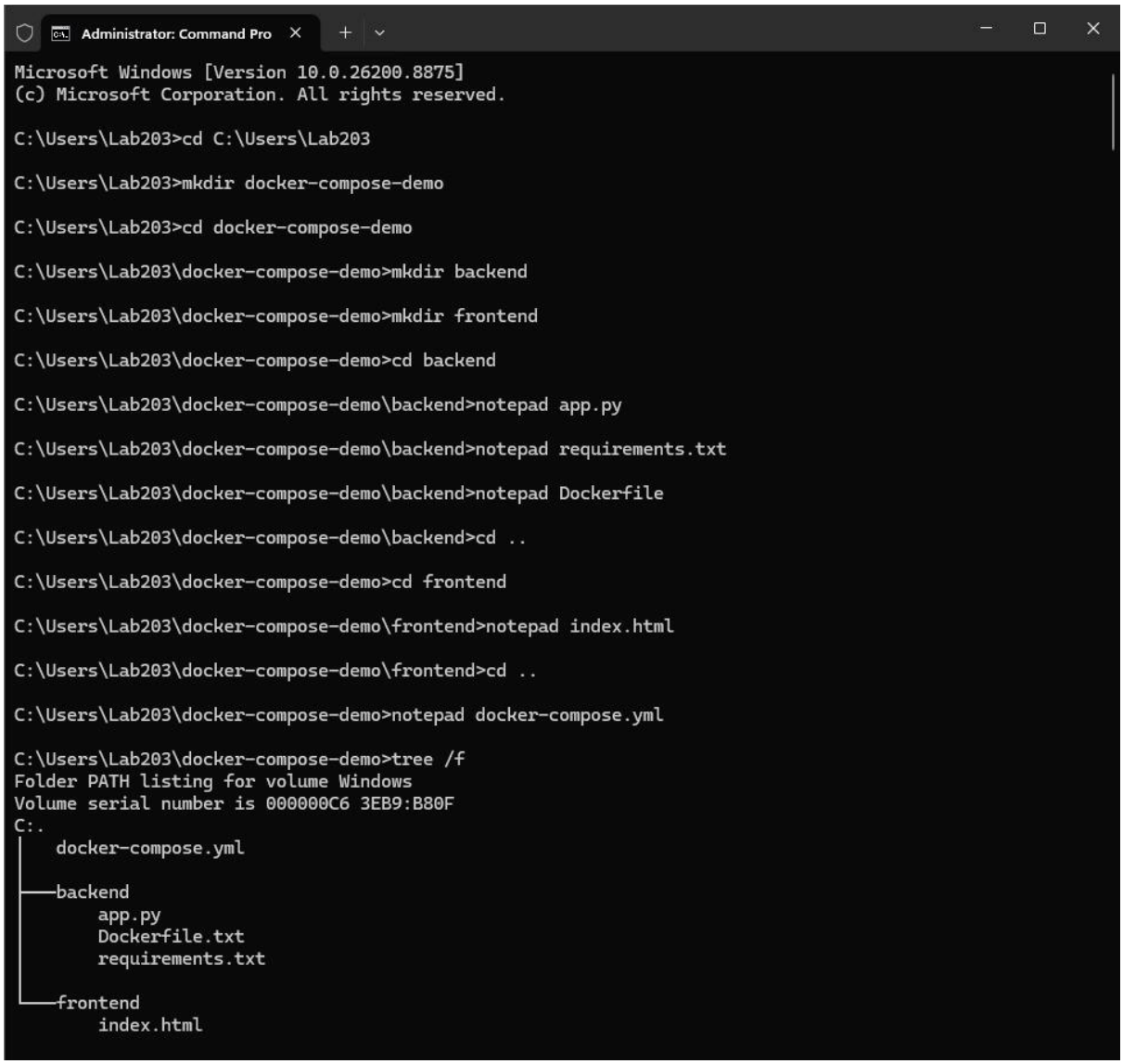
```
        proxy_pass http://web:5000;
    }

    location /health {
        proxy_pass http://web:5000;
    }
}
```

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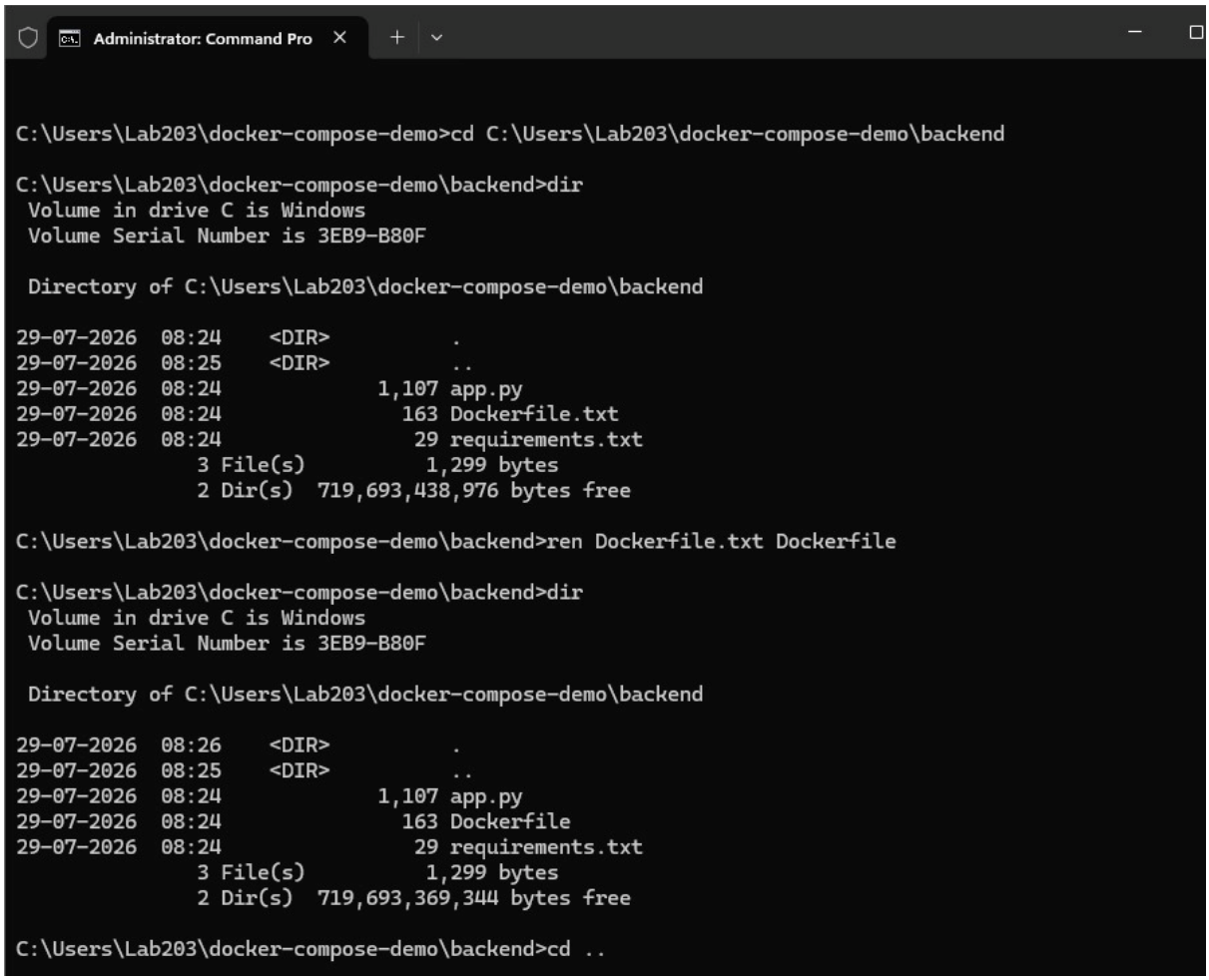
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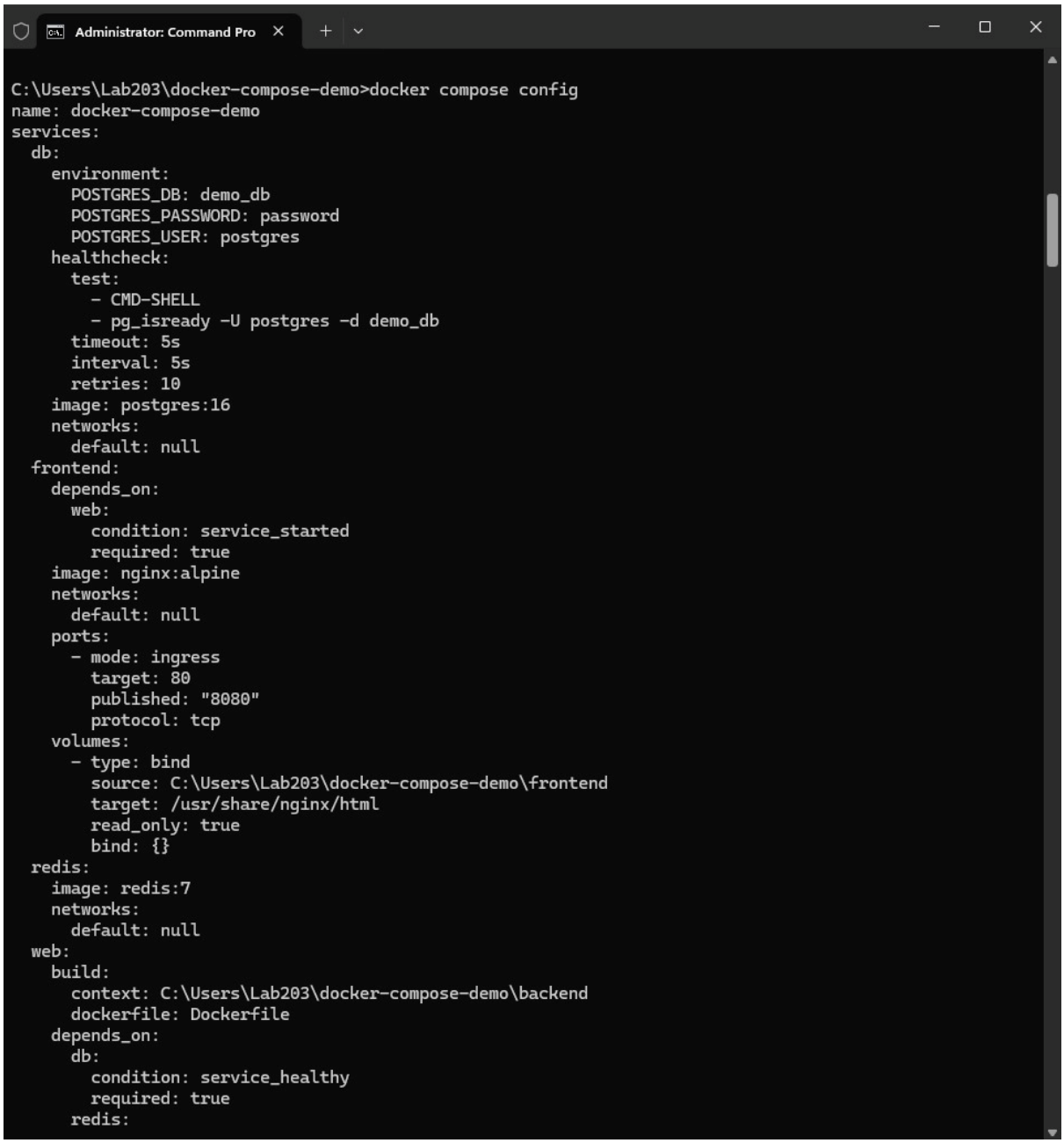
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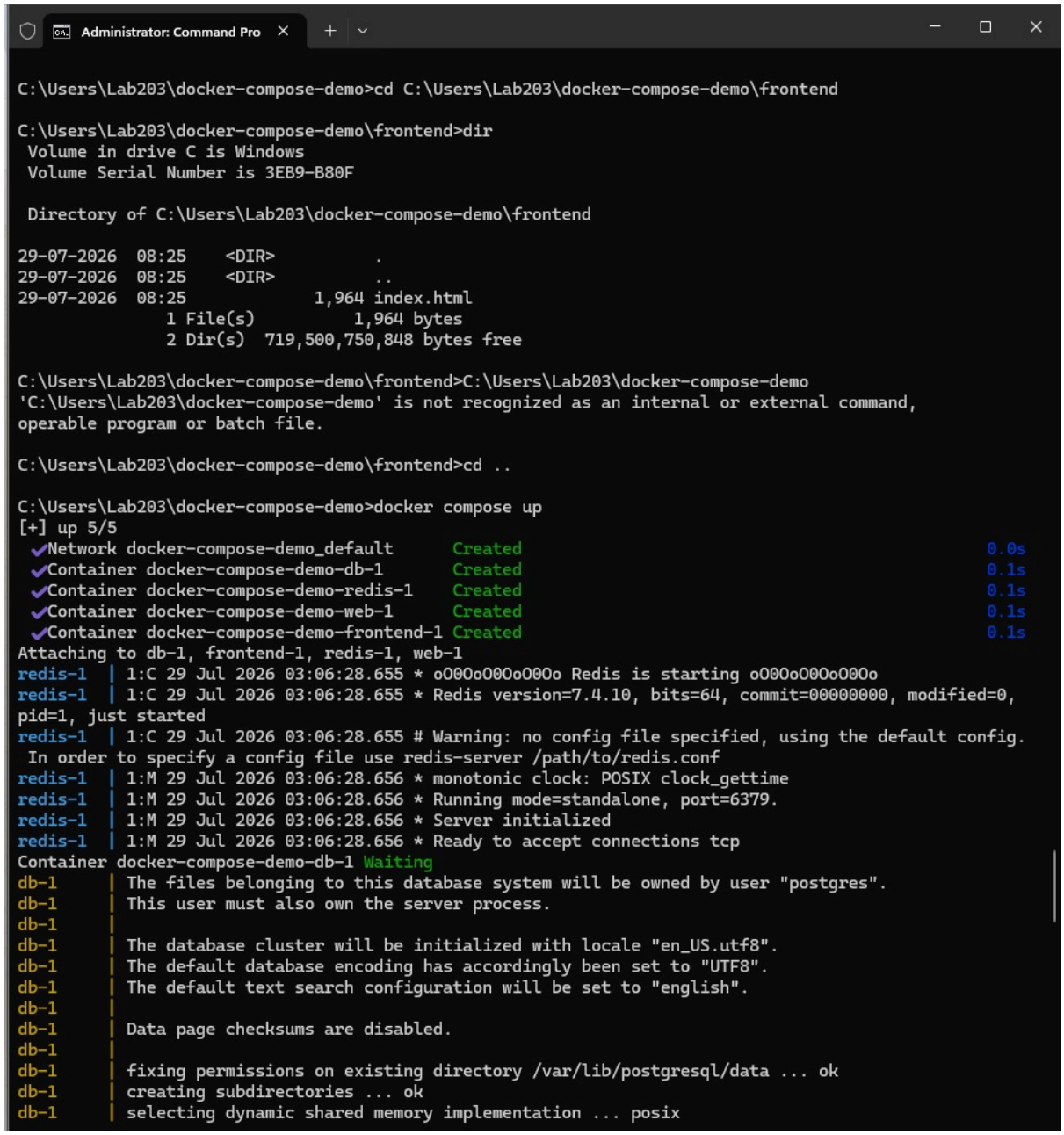
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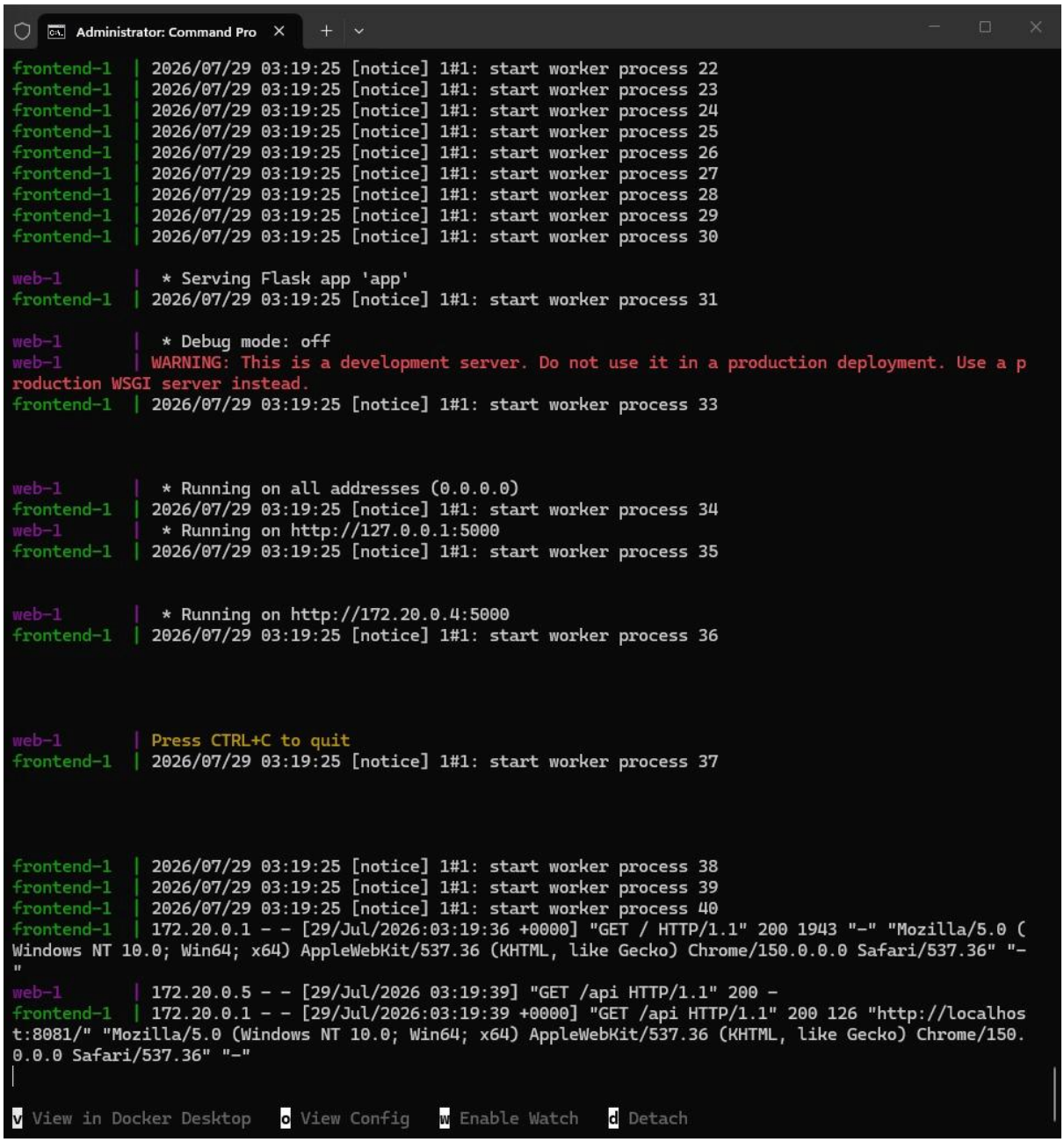
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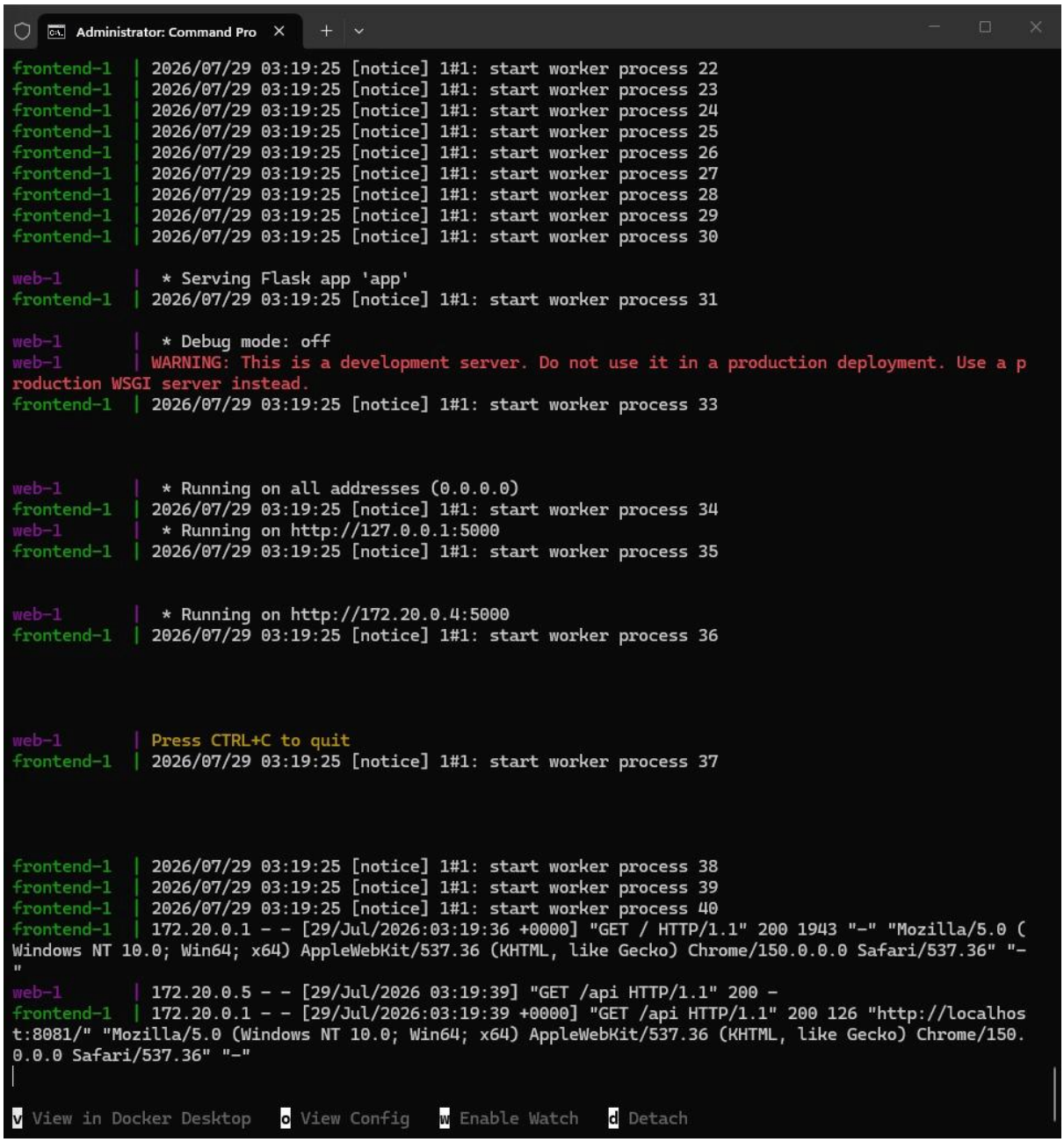
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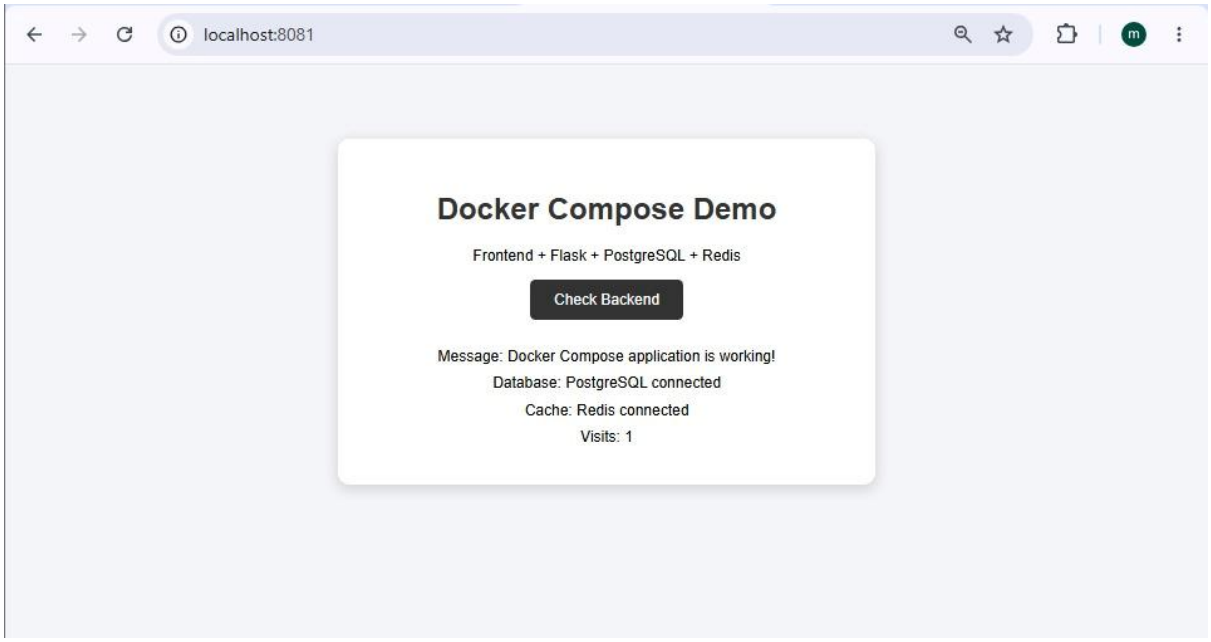
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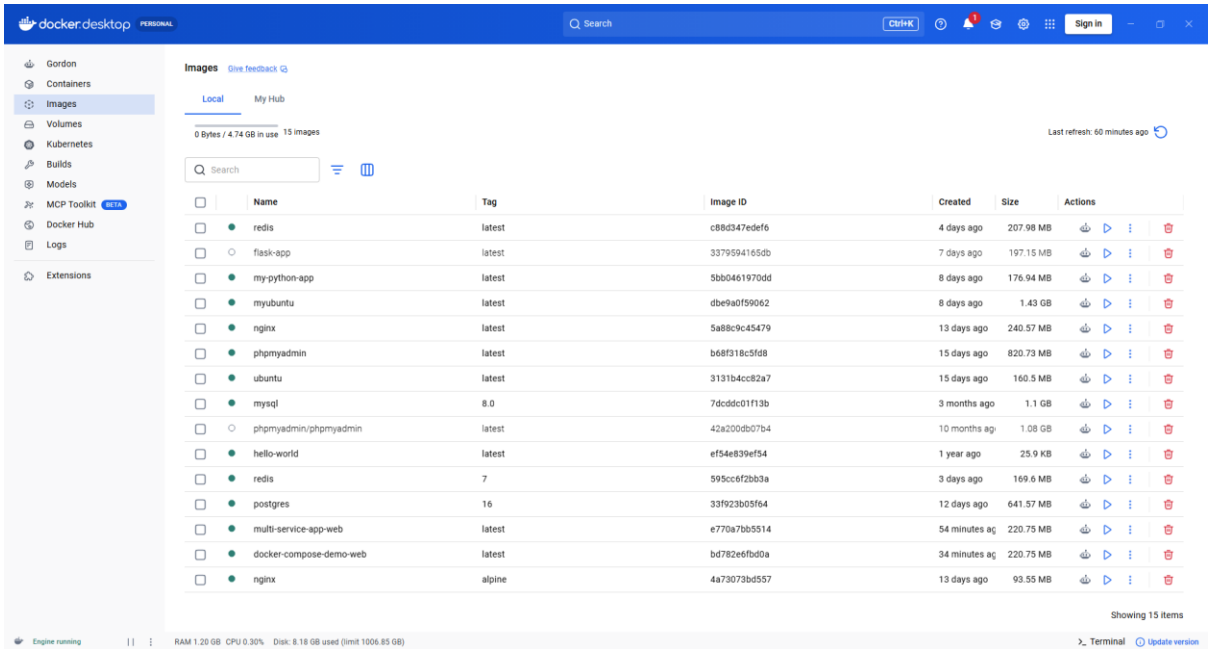


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Conclusion:

This experiment provided hands-on experience deploying a multi-service application using Docker and Docker Compose. Key takeaways include writing custom Dockerfiles, building container images, and orchestrating services—including an Nginx frontend, a Flask backend, a PostgreSQL database, and a Redis cache. Additionally, I gained practical knowledge in configuring inter-service communication and implementing health checks to ensure proper startup sequencing, highlighting how Docker Compose simplifies the management of complex application stacks.